

PROJECT APEX

Enterprise LNG Operations Intelligence Platform

VOLUME 4

Dashboard Design & Storytelling

Wireframes, UI/UX Guidelines, Navigation Design & Presentation Storytelling

PART OF THE SERIES

Atlantic LNG Operations Limited (ALOL) — Power BI Project

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Series Overview

Volume 1 — Project Planning & Business Understanding

Volume 2 — Data Preparation & Data Modeling

Volume 3 — DAX Development

Volume 4 — Dashboard Design & Storytelling (this document)

Volume 5 — Portfolio, Documentation & Interview Preparation

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1. Introduction & Objectives of This Volume

Volume 3 delivered a complete, tested DAX measure library — 45+ measures across seven subject areas with a master KPI definitions table. Volume 4 is where all of that analytical work becomes something a stakeholder can actually see, interact with, and make decisions from. This volume defines the visual product: what each of the nine dashboard pages looks like, how they relate to each other, and how to tell a coherent business story through the finished Power BI file.

This volume is written in two halves. The first half (Sections 2–5) covers the UI/UX system — the design rules, colour palette, typography, and navigation logic that apply consistently across every page. The second half (Sections 6–14) covers each dashboard page individually, with a wireframe diagram, the specific measures and visuals it uses, and the story each page is designed to tell.

1.1 The Nine Dashboard Pages

Page	Purpose	Primary Stakeholder
1. Executive Dashboard	Single-screen health summary of the entire business	CEO, Board, Managing Director
2. Production Dashboard	Train-level and product-level production deep-dive	Production Manager, Plant GM
3. Export Dashboard	Commercial performance, customers, and trade flow	Commercial Manager, CFO
4. Finance Dashboard	Revenue, cost, and profitability analysis	CFO, Finance team
5. Maintenance Dashboard	Work order tracking and equipment reliability	Maintenance Manager
6. Shutdown Dashboard	Shutdown events, duration, and cost analysis	Plant GM, Maintenance Manager
7. HSE Dashboard	Safety incident trends and severity analysis	HSE Manager, Regulator
8. Weather Dashboard	Bonny Island weather and marine conditions	Logistics, Shipping Coordinator
9. Map Dashboard	Geographic export flow and global trade patterns	Commercial Manager, CEO

2. Colour Palette

The ALOL Project APEX colour palette is built on a deep navy and professional blue foundation — chosen to convey the precision, trust, and operational seriousness expected of an enterprise energy analytics product. The palette is compact by design: four primary colours, three status colours, and one neutral range. Using fewer colours consistently is always more effective than using many colours inconsistently.

2.1 Primary Palette

Role	Colour Name	Hex Code	Used For
Primary Dark	Deep Navy	#1B2631	Page background, header bars, navigation strip
Primary Mid	ALOL Blue	#1B4F72	Section headers, table header rows, main KPI card backgrounds
Primary Light	Steel Blue	#2E86C1	Chart fill (primary series), accent lines, hover states
Accent	Teal Highlight	#117A65	Positive trend indicators, dimension table colour coding

2.2 Status / Traffic Light Colours

Status	Hex Code	Used For
Good / On-target	#1E8449 (Dark Green)	Positive YoY, availability above target, below-target incident counts
Warning / Near-threshold	#B7950B (Gold)	Metrics approaching but not breaching target thresholds
Alert / Off-target	#922B21 (Dark Red)	Negative YoY, availability below threshold, above-target incidents

2.3 Neutral / Background Range

Role	Hex Code	Used For
Canvas background	#F4F6F7	Report page canvas colour behind all visuals
Card background	#FFFFFF	Individual KPI card and visual backgrounds
Subtle fill	#EBF5FB	Alternate row shading in tables
Text — primary	#1B2631	All body text, data labels, axis labels
Text — secondary	#5D6D7E	Subtitle text, legend labels, helper text

2.4 Applying the Palette in Power BI

- Create a custom theme JSON file with these hex values and import it via View → Themes → Browse for themes — this applies the palette consistently across all visuals without manually recolouring each one
- All chart series should use Steel Blue (#2E86C1) as the default single-series colour, and the full primary palette sequence for multi-series charts
- KPI cards should use a white (FFFFFF) card background with ALOL Blue (#1B4F72) title text — never place coloured backgrounds on KPI cards unless using traffic-light conditional formatting
- Conditional formatting on KPI cards and tables should use the three status colours exclusively — never introduce additional colours for conditional states
- Dark backgrounds (Deep Navy #1B2631) should be used only for the navigation strip and page headers, not for chart backgrounds or card backgrounds, to maintain readability

3. Typography & Text Standards

Element	Font	Size	Weight	Colour
Page title (header bar)	Segoe UI	20pt	Bold	#FFFFFF on Navy background
Section heading	Segoe UI	14pt	Semi-bold	#1B4F72 (ALOL Blue)
KPI card value	Segoe UI	28–36pt	Bold	#1B2631 (Deep Navy)
KPI card label	Segoe UI	10pt	Regular	#5D6D7E (Secondary text)
Chart axis labels	Segoe UI	9pt	Regular	#5D6D7E
Chart data labels	Segoe UI	10pt	Regular	#1B2631
Table header text	Segoe UI	10pt	Bold	#FFFFFF on Blue
Table body text	Segoe UI	10pt	Regular	#1B2631
Tooltip title	Segoe UI	11pt	Bold	#1B4F72
Tooltip body	Segoe UI	10pt	Regular	#212121
Navigation button text	Segoe UI	9pt	Regular	#BDC3C7
Subtitle / caption	Segoe UI	9pt	Italic	#5D6D7E

Segoe UI is the standard throughout because it is Power BI's native default font, renders cleanly at all sizes on screen, and is available on every Windows machine where Power BI Desktop runs — avoiding any font substitution issues when the file is opened by other users.

4. Layout Grid & UI/UX Rules

4.1 Standard Page Layout

Every dashboard page in Project APEX follows the same structural grid. Consistency in layout is not aesthetic preference — it reduces the cognitive load on stakeholders who switch between pages, because their eyes know where to look for the title, the KPI cards, the main chart, and the navigation bar without relearning the layout every time.

Zone	Position	Height	Contents
Header bar	Top	~60px	Page title, ALOL logo placeholder, date last refreshed
KPI card row	Below header	~100px	4–6 KPI cards spanning full page width
Main visual area	Centre	~320px	Primary chart(s) — left 60% for main chart, right 40% for secondary
Supporting visuals	Below main	~180px	Supplementary charts, small multiples, or tables
Navigation strip	Bottom	~45px	Page navigation buttons — one per dashboard page

4.2 KPI Card Design Rules

- Every KPI card shows: the metric value (large, bold), the metric name (below value, smaller), and a comparison indicator (YoY %, vs target, or trend arrow) — never just a number with no context
- Limit to a maximum of 6 KPI cards per page — more than 6 clutters the card row and reduces impact for each individual metric
- Use conditional formatting on the comparison indicator only, not on the main value — the main value is always in Deep Navy (#1B2631) regardless of whether it is good or bad
- Do not use the Power BI default KPI visual for most cards — a plain Card visual with a separate trend line overlaid as a sparkline gives more design control; the native KPI visual has limited formatting options

4.3 Visual Selection Rules

Data type	Recommended Visual	Avoid
Single metric vs target	Card + conditional icon	Gauge (wastes space, hard to read precisely)
Trend over time	Line chart	3D line chart, area chart with heavy fill
Part-to-whole	Donut chart (max 5 segments) or Stacked bar	Pie chart (hard to compare slices)

Data type	Recommended Visual	Avoid
Comparison across categories	Clustered bar or horizontal bar	3D bar, exploded charts
Ranking	Horizontal bar sorted descending	Unsorted bar charts
Geographic distribution	Map visual (Bing Maps or Azure Maps)	Symbol maps without proper lat/lon
Detailed data grid	Matrix or Table visual	More than 7 columns in a single table
Cross-highlight relationship	Scatter chart	Bubble chart (hard to read at scale)

4.4 Interaction & Filter Rules

- Every page must have at least one Year slicer and one key dimension slicer (train, product, region, or similar depending on context) — slicers should be positioned at the top-right of the main visual area, not at the far left where they interrupt the natural left-to-right reading flow
- Cross-filtering (clicking a visual to filter others) should be left at Power BI's default (enabled) — do not disable cross-filtering without a specific reason, since stakeholders frequently use it intuitively even when not instructed to
- Slicers should use dropdown style rather than list style for most dimensions to save vertical space, except for Year and Quarter which should use tile/button style for at-a-glance selection
- Sync slicers should be used for Year across all pages — once a year is selected, it should persist as the user navigates between dashboard pages rather than resetting to all-years on every page

5. Navigation, Drill-Through & Tooltip Design

5.1 Navigation Plan

All nine pages are connected via a persistent navigation strip at the bottom of every page. Each page has a navigation button for every other page. This mirrors the design pattern used in professional Power BI apps and allows any stakeholder to jump from any page to any other page in a single click, without using the Pages pane.

Navigation buttons are built using Power BI's Button feature (Insert → Button → Page Navigator or manually configured Blank buttons) with the following spec:

Button label	Target page	Icon suggestion
Executive	Page 1 — Executive Dashboard	Home icon
Production	Page 2 — Production Dashboard	Factory/train icon
Export	Page 3 — Export Dashboard	Ship icon
Finance	Page 4 — Finance Dashboard	Currency icon
Maintenance	Page 5 — Maintenance Dashboard	Wrench icon
Shutdowns	Page 6 — Shutdown Dashboard	Warning icon
HSE	Page 7 — HSE Dashboard	Shield icon
Weather	Page 8 — Weather Dashboard	Cloud icon
Map	Page 9 — Map Dashboard	Globe icon

The currently active page button should have a lighter background (Steel Blue #2E86C1) compared to inactive buttons (Deep Navy #1B2631), providing a clear visual indicator of which page is currently open without requiring the stakeholder to read the page title.

5.2 Drill-Through Design

Drill-through is configured on three pages, allowing a user to right-click a data point on a summary page and drill into detail for that specific item:

Drill-through target page	Triggered from	Drill-through field
Train Detail (sub-page of Production)	Any visual showing TrainName	dim_trains[TrainName]
Customer Detail (sub-page of Export)	Customer name in any Export visual	dim_customers[CustomerName]
Incident Detail (sub-page of HSE)	Incident type or area in HSE visuals	fact_hse_incidents[IncidentType]

Each drill-through target page should include a back-button (Insert → Button → Back) in the top-left so users can return to the page they came from. The drill-through filter pill automatically appears at the top of the detail page, showing which context is active (e.g. 'Train: Train 3').

5.3 Tooltip Design

Custom tooltip pages (built as low-height Power BI report pages with the Tooltip toggle enabled under Page information) should be created for the following visuals, to provide additional context without requiring navigation away from the current page:

Visual receiving tooltip	Tooltip content
Production line chart (by train)	Train name, Train capacity, Average availability for the hovered period, LNG produced for that period
Shipment map scatter (Map Dashboard)	Customer name, Country, Product, Cargo quantity, Revenue, Shipment status
Maintenance bar chart (by equipment)	Equipment name, Equipment type, Criticality, Total maintenance cost, Corrective vs Preventive split
HSE incident area chart	Incident count by severity for hovered period, highest-risk area for that period, total lost time hours
Shutdown timeline	Train, Shutdown type, Start date, End date, Duration, Estimated lost production

6. Page 1 — Executive Dashboard

The Executive Dashboard is the landing page of the entire report. It is the only page any senior executive may ever spend more than 60 seconds on, so it must communicate the state of the entire business in one glance — no scrolling, no drilling required for the top-level story.

6.1 Wireframe

PROJECT APEX — Executive Dashboard ALOL Last Refreshed: [Date]		
[Avg Availability %] [Total LNG Produced YTD] [Profit Margin %] [Unplanned Shutdown Share %] [Total HSE Incidents] [Export Revenue YoY %]		
LINE CHART LNG Production vs Revenue 2019–2024 (monthly trend) Dual axis: Production (left), Revenue (right)	BAR CHART Profit Margin % by Year Compare all 6 years Conditional: Green/Gold/Red	DONUT Shutdown Type Mix Planned vs Unplanned Current filter year
TABLE Train-by-Train Summary Availability Production Cost Sorted by Availability desc	SCATTER CHART HSE Incidents vs Maintenance Cost By Train Bubble = LNG Output	KPI CARDS YoY Changes Revenue Production HSE
[Executive] [Production] [Export] [Finance] [Maintenance] [Shutdowns] [HSE] [Weather] [Map]		

6.2 KPI Cards — Executive Page

Card	Measure (Volume 3 ref)	Conditional Format Rule
Avg Plant Availability %	Average Availability %	Green >90%, Gold 80–90%, Red <80%
Total LNG Produced YTD	LNG Produced YTD	No conditional — informational
Profit Margin %	Profit Margin %	Green >30%, Gold 20–30%, Red <20%
Unplanned Shutdown Share %	Unplanned Shutdown Share %	Green <30%, Gold 30–50%, Red >50%
Total HSE Incidents	Total HSE Incidents	Green <50, Gold 50–100, Red >100 (note reversed convention)
Export Revenue YoY %	Export Revenue YoY %	Green positive, Red negative

6.3 Story This Page Tells

“In one screen, the CEO sees whether the plant is running reliably, whether it is making money, whether safety is trending in the right direction, and whether this year is better or worse than last year. Every other dashboard page in the report is a drill-down into one of those four questions.”

7. Page 2 — Production Dashboard

The Production Dashboard is the most data-dense page in the report. It is the daily working tool for the Production Manager and Plant GM, and must support train-level comparison, daily/monthly trend analysis, and product mix visibility in a single scrollable-free layout.

7.1 Wireframe

PROJECT APEX — Production Dashboard Train: [All] Year: [All]		
[Total LNG Produced] [Avg Availability %] [Avg Utilization %] [Conversion Efficiency %] [Production YoY %]		
LINE CHART Monthly LNG Production by Train 6 coloured series, one per train Slice by Year slicer	STACKED BAR Product Mix (LNG / LPG / Condensate) By Month, current year Shows proportional output	TRAIN SLICER [Train 1][Train 2][Train 3][Train 4][Train 5][Train 6] Button-style tile slicer
BAR CHART (horizontal) Train Performance Ranking Sorted by LNG Produced Conditional bar colour by Availability	AREA CHART Availability % vs Utilization % Over time, current filter context Dual shaded areas	MATRIX TABLE Train x Month Production Grid Heatmap conditional formatting Green=high, Red=low output
[Executive] [Production] [Export] [Finance] [Maintenance] [Shutdowns] [HSE] [Weather] [Map]		

7.2 Key Design Decisions

- The monthly production line chart uses all six train series simultaneously — use the dim_trains colour assignment to give each train a consistent colour across all pages of the report, not just this one, so that ‘Train 3 = orange’ is a consistent visual shorthand throughout the entire file
- The heatmap matrix (Train x Month) is the signature visual on this page — it immediately shows which trains struggled in which months without requiring any interaction, and is often the first visual executives point to when asking ‘what happened to production in Q3?’
- Both the Availability and Utilization measures from Volume 3 should appear on the same area chart with different shading, since the gap between them (available but not fully utilized) is itself analytically meaningful

7.3 Drill-Through from This Page

Clicking on any train name in the Train Performance Ranking bar chart should trigger the drill-through to the Train Detail sub-page, showing a full operational history for that specific train including its shutdown events, maintenance cost, and HSE incidents — all filtered to that single train. This is configured in Power BI via the Drill-through field well on the detail page, as specified in Section 5.2.

8. Page 3 — Export Dashboard

The Export Dashboard serves the commercial team and the CFO, showing who is buying ALOL's products, where they are shipping to, and whether revenue is growing, contracting, or shifting between customers and regions.

8.1 Wireframe

PROJECT APEX — Export Dashboard Product: [All] Year: [All]		
[Total Export Revenue] [Total Shipments] [Distinct Customers Served] [Avg Price per Unit] [Customer Concentration % Top 5] [Export Revenue YoY %]		
BAR CHART Revenue by Customer (Top 10) Horizontal, sorted descending Colour by ContractType	LINE CHART Monthly Shipment Revenue Trend Compare current vs prior year Dual series with shading	DONUT CHART Revenue by Product Type LNG / LPG / Condensate / NGL With % data labels
BAR CHART Shipments by Region Europe / Asia / Americas / etc Grouped by ShipmentStatus	TABLE Top 20 Shipments by Revenue Date Customer Product Qty Price Revenue Sortable columns	PRODUCT SLICER [LNG][LPG Propane][LPG Butane] [Condensate][NGL] Button-style tile slicer
[Executive] [Production] [Export] [Finance] [Maintenance] [Shutdowns] [HSE] [Weather] [Map]		

8.2 Key Design Decisions

- The Customer Concentration % (Top 5) KPI card from Volume 3 should be conditionally formatted in reverse — red when concentration is high (>70%), since high concentration is a commercial risk, not a positive
- The Revenue by Customer bar chart should only show customers with ShipmentStatus 'Delivered' by default, using a visual-level filter, to show realized rather than in-transit or cancelled revenue — add a toggle button to show all statuses for the commercial manager's use
- The Distinct Customers Served KPI (Volume 3, Section 4) is specifically useful here as a market breadth indicator — a declining trend signals commercial risk even if total revenue is stable

9. Page 4 — Finance Dashboard

The Finance Dashboard translates operational performance into financial outcomes. It is the CFO’s page — built around revenue, profitability, cost structure, and trend versus prior year.

9.1 Wireframe

PROJECT APEX — Finance Dashboard Year: [All] Product: [All]		
[Total Revenue] [Total OPEX] [Total CAPEX] [Total Profit] [Profit Margin %] [Profit YoY %]		
CLUSTERED BAR Revenue vs OPEX vs CAPEX by Year Grouped bars, 3 series Shows cost structure evolution	LINE CHART Profit Margin % Trend by Month Current year vs prior year Target line at 30% margin	BAR CHART Revenue per Tonne LNG by Year Shows pricing and efficiency trend Single series, Steel Blue
WATERFALL CHART Revenue to Profit Bridge Revenue → minus OPEX → minus CAPEX → minus Maintenance → = Profit	TABLE Financial Summary by Product Revenue OPEX CAPEX Profit Margin% For all 7 product types	AREA CHART Profit YTD Cumulative Current year running total Shaded area fill
[Executive] [Production] [Export] [Finance] [Maintenance] [Shutdowns] [HSE] [Weather] [Map]		

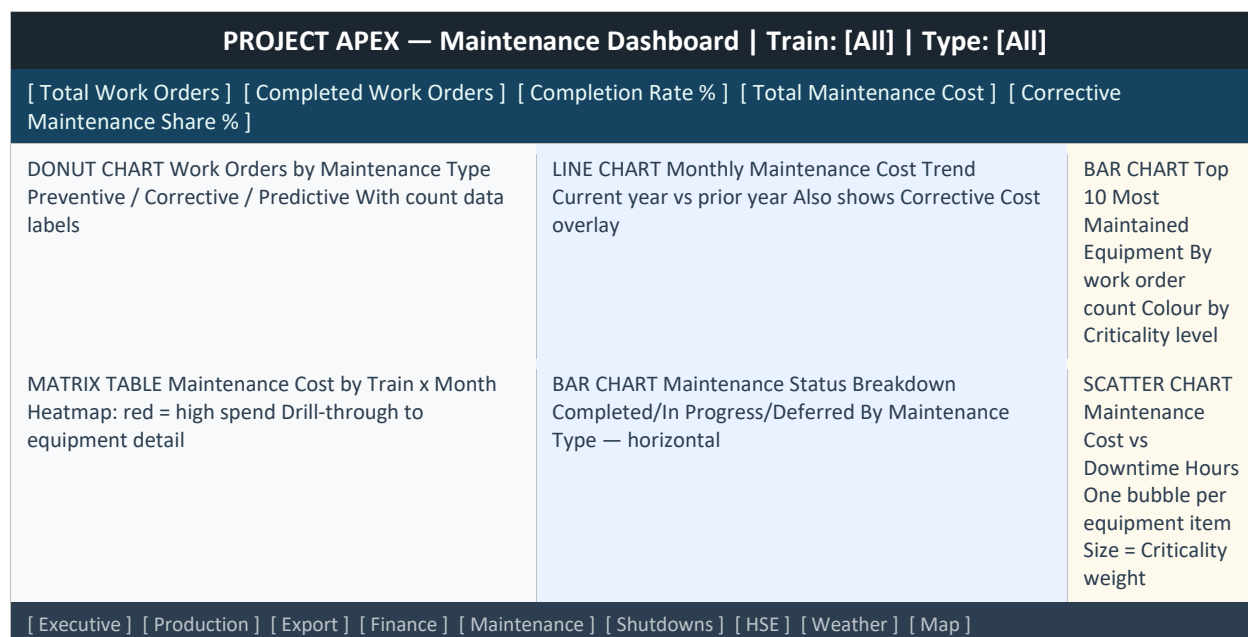
9.2 Key Design Decisions

- The Waterfall chart (Revenue-to-Profit bridge) is the signature visual on this page — it is the single most immediately understood profitability visual for non-finance executives, and requires no chart literacy to read. Build it using Power BI’s native Waterfall visual with Revenue as the start value and each cost category as a Breakdown field
- Place a constant reference line at the 30% Profit Margin target on the Profit Margin % trend line — this makes deviation from target immediately obvious without requiring the viewer to remember the target value while reading the chart
- The Revenue per Tonne LNG measure from Volume 3 (Section 8) is the cross-subject-area measure that links fact_finance and fact_production — explain this linkage in the visual subtitle so the CFO understands they are seeing both commercial pricing and operational output combined in one figure

10. Page 5 — Maintenance Dashboard

The Maintenance Dashboard is the Maintenance Manager's operational tool, tracking the work order backlog, cost trends, equipment reliability, and the critical ratio between corrective and preventive maintenance spend.

10.1 Wireframe



10.2 Key Design Decisions

- The Corrective Maintenance Share % KPI card should be the most prominently coloured card on this page — it is the single most important maintenance health indicator (Section 5 of Volume 3 explains why), and should use the red/gold/green traffic-light conditional formatting with red at >50% corrective share
- The scatter chart (Maintenance Cost vs Downtime Hours, sized by criticality) is designed to surface the equipment items that simultaneously cost the most to maintain AND cause the most downtime — those are the candidates for equipment replacement or upgrade conversations
- The Equipment Reliability Score measure from Volume 3 Advanced DAX section should appear as a column in the Maintenance Status table, giving a composite view of each equipment item's health alongside its raw cost and work-order count

11. Page 6 — Shutdown Dashboard

The Shutdown Dashboard gives plant leadership a complete history of every planned and unplanned shutdown event, their duration, cost, and relationship to lost production. The USERELATIONSHIP() measures from Volume 3, Section 6 are specifically surfaced on this page.

11.1 Wireframe

PROJECT APEX — Shutdown Dashboard Train: [All] Type: [All]		
[Total Shutdown Events] [Planned Turnaround Count] [Emergency Shutdown Count] [Unplanned Share %] [Estimated Revenue Lost] [Avg Duration (Days)]		
GANTT-STYLE BAR CHART Shutdown Timeline by Train Each bar = one shutdown event Colour by ShutdownType	CLUSTERED BAR Shutdown Events by Year & Type Grouped: Planned / Emergency / Other Trend of unplanned rate visible	DONUT CHART Shutdown Type Distribution All 4 types Current filter With count and % labels
TABLE All Shutdown Events Detail Train Type Start End Days Lost Prod Cost Sortable, filterable	BAR CHART Avg Shutdown Duration by Type Compare Planned vs Emergency With target duration lines	KPI CARDS Ended in Period Count Uses USERELATIONSHIP() measure Vs Started in Period
[Executive] [Production] [Export] [Finance] [Maintenance] [Shutdowns] [HSE] [Weather] [Map]		

11.2 Key Design Decisions

- The Gantt-style timeline is the signature visual on this page. Power BI does not have a native Gantt chart, so use the custom 'Gantt' visual from the AppSource marketplace, or simulate it using a Range Bar chart in a custom visual. Each bar spans StartDate to EndDate, and is coloured by ShutdownType — this makes overlapping shutdowns across trains immediately visible
- The 'Shutdown Events Ended in Period' KPI card is the one place in the entire report where the inactive EndDateKey relationship is activated via USERELATIONSHIP() — place a small footnote beneath this card explaining that it filters by end date rather than start date, so stakeholders do not assume it is a duplicate of the events-started count
- The Estimated Revenue Lost to Shutdowns measure (Volume 3, Section 10 — Advanced DAX) should appear on this page as a financial consequence KPI, connecting operational events directly to financial impact and completing the loop back to the Executive Dashboard's profitability picture

12. Page 7 — HSE Dashboard

The HSE Dashboard is the safety record of the company. It must meet two simultaneous requirements: being visually clear for senior leadership who need a trend at a glance, and being detailed enough for the HSE Manager to identify intervention priorities by area, equipment type, and incident severity.

12.1 Wireframe

PROJECT APEX — HSE Dashboard Severity: [All] Year: [All]		
[Total HSE Incidents] [Critical Incidents] [Total Lost Time Hours] [LTIFR] [HSE Incident YoY %]		
STACKED BAR Incidents by Severity per Month Low/Medium/High/Critical stacked Conditional colour per severity	LINE CHART Cumulative Incidents YTD Current year vs prior year Line with filled area gap	DONUT CHART Incidents by Type Near Miss / LTI / Fire / Spill etc Top 5 types visible
BAR CHART (horizontal) Top 10 Areas by Incident Count Colour by avg severity weight Highest-risk areas at top	MATRIX TABLE Incident Heatmap: Area x Month Cell = incident count Conditional: white to dark red	TABLE High & Critical Incidents Detail Date Area Type Lost Hrs Employee Filtered to High+Critical only
[Executive] [Production] [Export] [Finance] [Maintenance] [Shutdowns] [HSE] [Weather] [Map]		

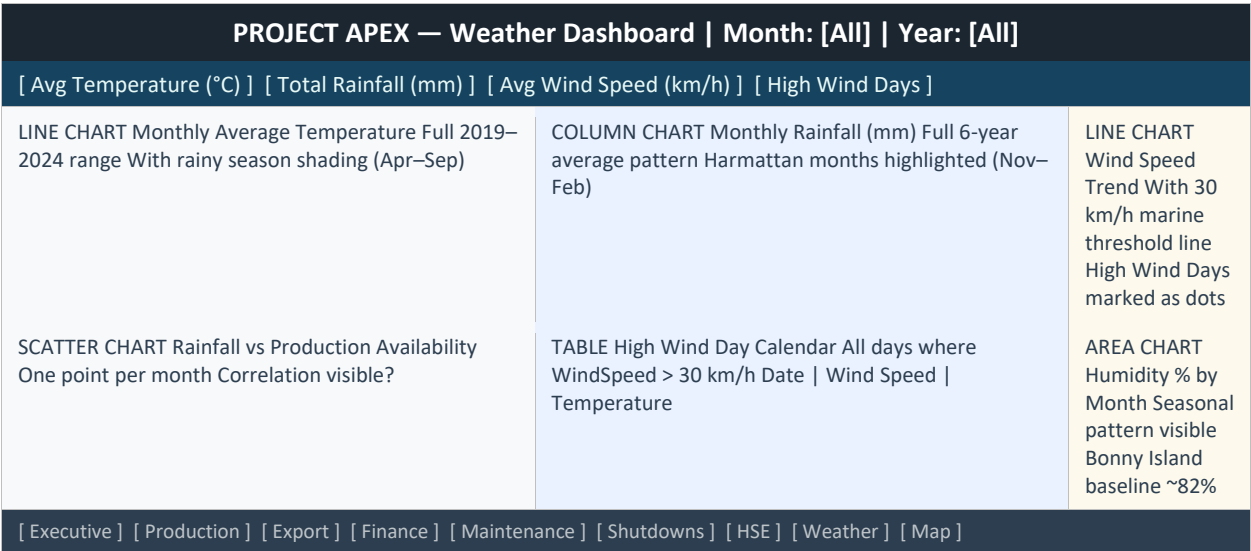
12.2 Key Design Decisions

- The YoY% conditional formatting on this page must be reversed compared to every other page in the report — down-arrows should be green (fewer incidents = better) and up-arrows should be red. This is the single most common HSE dashboard design mistake — using the default ‘up = good’ convention on a safety metric produces a dashboard that shows a worsening safety record as a positive trend
- The SeverityWeight Power Query column created in Volume 2 (Section 6.3) is used here to colour the Top 10 Areas bar chart by average severity weight rather than incident count — a high-count area with mostly Near Misses may be less concerning than a lower-count area with multiple High-severity events
- The Incident Detail table at the bottom-right should be filtered at the visual level to show only High and Critical severity incidents — this gives the HSE Manager immediate access to the records that require follow-up action, without scrolling through hundreds of Near Miss records to find them

13. Page 8 — Weather Dashboard

The Weather Dashboard connects Bonny Island’s climate data to operational and shipping context. It is the smallest subject-area page in the report but provides important supporting insight for both production and logistics decisions.

13.1 Wireframe



13.2 Key Design Decisions

- The rainy season shading (April to September) on the temperature and rainfall charts is added as a reference band or constant line pair, not as a data series — use a reference line set to the month range to add a translucent blue background band that visually marks the rainy season without adding noise to the underlying data
- The scatter chart correlating Rainfall with Production Availability is deliberately exploratory — it may or may not show a strong correlation in this synthetic dataset, and that is fine. The point is to place the question visibly in the report so stakeholders can see whether the relationship is real, rather than assuming it either is or is not without looking
- The 30 km/h marine threshold reference line on the wind chart is the most practically important visual element on this page for the Logistics team — it directly answers the question ‘how many days per year is the jetty potentially unsafe for loading operations’

14. Page 9 — Map Dashboard

The Map Dashboard is the geographic view of ALOL’s global export business. It uses the Latitude and Longitude columns in dim_countries (confirmed in Volume 1, Section 8.1) to place every export destination on a world map, sized or coloured by revenue, volume, or shipment count depending on the active measure selection.

14.1 Wireframe

PROJECT APEX — Map Dashboard Product: [All] Year: [All]		
[Total Export Revenue] [Countries Reached] [Top Region by Revenue] [Total Shipments]		
WORLD MAP (full width) Bubble map via Azure Maps or Bing Maps One bubble per destination country Bubble size = Revenue Colour = Region Tooltip: Customer Product Revenue Shipments		REGION SLICER [Europe][Asia][Americas] [Middle East][Africa] Button tile slicer (right panel)
BAR CHART Revenue by Region Horizontal sorted desc Europe / Asia / Middle East / Americas / Africa	TABLE Country Performance Country Region Revenue Shipments Top Product Top 20 by Revenue	DONUT Region Revenue Split 5 regions as donut slices Labels show % and absolute
[Executive] [Production] [Export] [Finance] [Maintenance] [Shutdowns] [HSE] [Weather] [Map]		

14.2 Building the Map Visual

1. Use the Azure Maps visual (available natively in Power BI without a custom visual import) or the standard Power BI Map visual with Bing Maps.
2. In the Location field well, place dim_countries[CountryName]. Power BI will attempt automatic geocoding, but since dim_countries already has Latitude and Longitude columns (from Volume 1 design), use those instead: place CountryName in Location, Latitude in Latitude, and Longitude in Longitude — this guarantees accurate placement for all 60 countries, even unusual ones that Bing’s geocoder might misplace.
3. Place [Total Export Revenue] in Size to scale bubbles by revenue. Place dim_countries[Region] in Legend to colour bubbles by region.
4. Enable the custom tooltip defined in Section 5.3 on this visual so hovering over any country bubble shows Customer name, Product, Cargo quantity, Revenue, and Shipment status.
5. Set a visual-level filter to exclude Nigeria (CountryID = 41) from the map if ALOL’s domestic gas deliveries are being counted as ‘export’ in the dataset — domestic deliveries should not appear as export destinations on a global trade map.

14.3 Key Design Decisions

- The Map Dashboard should use a dark map background (Azure Maps' dark style or the Road-Dark Bing Maps option) to match the Deep Navy (#1B2631) report theme — a white map background clashes with every other page in the report
- The Countries Reached KPI card should use a DISTINCTCOUNT of DestinationCountryID from fact_export_shipments (with Delivered status filter) — not the total 60 rows of dim_countries, which would always return 60 regardless of actual trading activity

15. Storytelling Guide

A technically correct dashboard that cannot be narrated is not a finished product. This section provides a storytelling guide for presenting the Project APEX dashboard suite to an audience — whether that is a mock stakeholder presentation for a portfolio review, a job interview demonstration, or a real business briefing.

15.1 The Three-Act Structure

Structure any presentation of this dashboard in three acts, regardless of how much time you have:

6. Act 1 — The State of the Business (Executive Dashboard only, 2–3 minutes): Start here. State the headline — whether the plant is up or down on availability, whether the company made more or less money this year, and whether safety improved or worsened. Let the Executive Dashboard do the talking; your job in Act 1 is simply to narrate the KPI cards.
7. Act 2 — The Story Behind the Headlines (Production, Export, and Finance dashboards, 5–7 minutes): Choose the one or two numbers from Act 1 that were most interesting (the best or worst) and drill into them. If availability was the lowest it has been in three years, open the Production Dashboard and show which train drove it. If profit margin improved, open the Finance Dashboard and show whether that was revenue growth or cost reduction.
8. Act 3 — Risks and Opportunities (Maintenance, Shutdown, HSE dashboards, 3–5 minutes): Close with what the data says about forward-looking risk. High corrective maintenance share on Train 3? A rising unplanned shutdown rate? An HSE incident spike in Process Area B? These are the conversations that move from ‘reporting what happened’ to ‘informing what to do next.’

15.2 Opening Line Suggestions

“Project APEX is an enterprise analytics platform built to give Atlantic LNG Operations Limited a single source of operational and commercial truth across six years of simulated operations. What I want to show you today is not just that the dashboards look good — it is that every number on every page can be traced back to a business decision someone in this company needs to make.”

15.3 Handling Questions During a Demo

- If asked ‘why did you choose this visual type?’ — refer back to Section 4.3 (Visual Selection Rules) and explain the specific reason for that visual in that context
- If asked ‘how do you know this DAX measure is correct?’ — refer back to Volume 3, Section 12 (Validation Checklist) and describe the spot-check process used
- If asked ‘what would you do differently with more time?’ — good answers include: adding a What-If parameter for price sensitivity analysis on the Finance page, incorporating actual hours-worked data to make the LTIFR measure precise, and adding anomaly detection using Power BI’s Smart Narratives for the Production trend

- If asked ‘how does this relate to your real-world skills?’ — frame every technical decision in terms of the business problem it solves (Section 4 of Volume 1), not in terms of the tool or formula used

What Comes Next: Volume 5 Preview

Volume 4 has defined the complete visual product — nine dashboard pages, a full UI/UX system, drill-through and tooltip specifications, and a storytelling guide. Volume 5, the final volume, packages all of that into portfolio-ready deliverables: a LinkedIn portfolio description, a GitHub README, a presentation script, and a full interview preparation guide with likely questions and model answers drawn directly from the technical decisions made across Volumes 1–4.

End of Volume 4 — Dashboard Design & Storytelling